1. A council of size 14 is to be selected from 10 women and 10 men. How many different councils can be selected if:

   a) There is no restriction?

   b) There must be equal number of women and men?

   c) There must be between 4 and 6 men?

2. Consider alphabet 0, 1, 2, 3, 4, 5, 6, 7, 8 and strings of length 10.

    a) How many strings have exactly three 4's?

    b) How many strings have exactly two 4's and five 3's?

    c) How many strings have exactly six 1's and no other digits repeated?

    d) How many strings have at least seven 2's?

    e) How many strings with exactly three 0's have no adjacent 0's?

3. What value does a variable *counter* have after the following code execution?

a

```
counter = 100

for i = 1 to 5 {
        counter = counter + 7
        for j = i+1 to 5 {
                counter = counter + 8
        }
}
```

b

```
counter = 100

for i = 1 to 10 {
        counter = counter + 7
        for j = i+1 to 10 {
                counter = counter + 8
                for k = j+1 to 10 {
                        counter = counter + 9
                }
        }
}
```

c

```
counter = 100

for i = 1 to 500 {
        counter = counter + 7
        for j = i+1 to 400 {
                counter = counter + 8
        }
}
```

d

```
counter = 100

for i = 1 to 400 {
        counter = counter + 7
        for j = i+1 to 500 {
                counter = counter + 8
        }
}
```

e

```
counter = 100
a = − 300
for i = 1 to n {
        counter = counter + 7
        a = i
}
for j = a to 2n {
        counter = counter + 8
        for k = j+1 to 2n {
                counter = counter + 9
        }
}
// assume n>1
```

Review lecture topic “Combinations with repetition allowed”. Then do the exercises:

4. In how many ways can a student distribute 10 identical marbles into six distinct containers such that:

    a) Every container has at least one marble?

    b) The last container has 5 or more odd number of marbles.

5. In how many ways can a student arrange 10 distinct books on three shelves? Shelves could be empty. Books are placed next to each other and left to right with no empty spots between them.

If all of the problems can’t be done in the lab period, students can work on them on their own.